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LEARNING CONTROL DEVICE, LEARNING CONTROL METHOD, AND MAGNETIC DISK DEVICE

Abstract

A learning-control device includes a feedback-control unit **30** and a learning-control unit **40**. The feedback-control unit **30** outputs, based on an input signal according to a tracking error between an operation-result-state of a control target operating according to an input-control-signal based on a feedback-signal and a target-state, the feedback-signal causing the operation-result-state of the control target **34** to track the target-state. The learning-control unit **40** outputs to the feedback-path F, through which the input signal according to the tracking error is input to the feedback-control unit **30**, the learning-control input updated according to the tracking error causing the tracking error to approach zero asymptotically. The evaluation section length of an evaluation section by the learning-control unit **40** for the tracking error is longer than the output section length of the output section in which the learning-control unit **40** outputs the learning-control inputs to the feedback-path F.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-032652, filed on Mar. 5, 2024; the entire contents of which are incorporated herein by reference.
FIELD
[0002] Embodiments of the present disclosure described herein relate to a learning control device, a learning control method, and a magnetic disk device.
BACKGROUND

[0003] As a digital control device, a learning control device is known that iteratively controls a control target according to learning control inputs stored in a learning memory and uses a tracking error between a target value and an output value of the control target to sequentially update the learning control inputs to be used for a next iterative learning to improve a control performance.
[0004] However, by conventional technologies, although the tracking error is suppressed and an accuracy of an operation result state of the control target is improved while the learning control inputs updated by learning control and the target value are applied to a feedback path, a transient response occurs at the moment when the learning control is terminated, and the accuracy of the operation result state of the control target relative to a target state may deteriorate.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0005] FIG. **1** is a block diagram of a configuration of a main part of a hard disk drive (HDD);
[0006] FIG. **2** is a block diagram of an example configuration of a seek control section;
[0007] FIG. **3** is an explanatory diagram of conventional learning control;
[0008] FIG. **4** is a schematic diagram of the seek control section;
[0009] FIG. **5** is an explanatory diagram of the conventional learning control;
[0010] FIG. **6** is an explanatory diagram of the conventional learning control;
[0011] FIG. **7** is a schematic diagram of a relationship between an evaluation section length and an output section length in the conventional technology;
[0012] FIG. **8** is a schematic diagram of a relationship between an evaluation section length and an output section length according to the present embodiment;
[0013] FIG. **9** is an explanatory diagram of a learning control;
[0014] FIG. **10** is a diagram illustrating an impulse response;
[0015] FIG. **11** is an explanatory diagram of a finite impulse response (FIR) filter;
[0016] FIG. **12** is a schematic diagram of a learning control unit;
[0017] FIG. **13** is an explanatory diagram of a result of a numerical simulation;
[0018] FIG. **14** is an explanatory diagram of a result of a numerical simulation;
[0019] FIG. **15** is an explanatory diagram of a result of a numerical simulation;
[0020] FIG. **16** is an explanatory diagram of a result of a numerical simulation;
[0021] FIG. **17** is an explanatory diagram of a result of a numerical simulation;
[0022] FIG. **18** is an explanatory diagram of results of numerical simulations; and
[0023] FIG. **19** is an explanatory diagram of results of numerical simulations.
DETAILED DESCRIPTION
[0024] According to an embodiment, a learning control device includes one or more hardware processors configured to function as a feedback control unit and a learning control unit. The feedback control unit outputs, based on an input signal according to a tracking error between an operation result state of a control target operating according to an input control signal based on a feedback signal and a target state, the feedback signal to cause the operation result state of the control target to track the target state. The learning control unit outputs to a feedback path, through which the input signal according to the tracking error is sent to be input to the feedback control unit, a learning control input updated according to the tracking error to cause the tracking error to approach zero asymptotically. The evaluation section length of an evaluation section by the learning control unit for the tracking error is longer than an output section length of an output section in which the learning control unit outputs the learning control input to the feedback path.
[0025] Embodiments of a learning control device, a learning control method, and a magnetic disk device according to the present disclosure will be described in detail below with reference to accompanying drawings. The present disclosure is not limited to the following embodiments.
[0026] FIG. **1** is a block diagram illustrating an example of a configuration of a main part of a hard disk drive (HDD) **10** according to the present embodiment. The HDD **10** is an example of the magnetic disk device and the learning control device.
[0027] A Host **20** is a device that uses the HDD **10** as a storage device. The Host **20** is connected to the HDD **10** by a host interface IF.
[0028] The HDD **10** includes one or more magnetic disks **11**, magnetic heads **12**, a spindle motor (SPM) **13**, micro actuators (MAs) **14**, support members **14***a*, a voice coil motor (VCM) **15**, an arm **15***a*, a driver integrated circuit (driver IC) **16**, a head IC **17**, and a system large scale integration (system LSI) **18**.
[0029] The magnetic disks **11** are magnetic recording media and are stacked and arranged at regular intervals. The magnetic disk **11** has an upper disk surface and a lower disk surface. In the present embodiment, both surfaces (upper disk surface and lower disk surface) of the magnetic disk **11** are recording surfaces on which data is magnetically recorded. The magnetic disk **11** is rotated at high speed by the SPM **13**. The HDD **10** may include one magnetic disk **11** or a plurality of (for example, 10 or more of) the magnetic disks **11**.
[0030] The magnetic heads **12** are disposed so as to correspond to the recording surfaces in both surfaces (upper disk surface and lower disk surface) of each magnetic disk **11** and are used to write data to the recording surface of the magnetic disk **11** and to read the data written in the recording surface. The magnetic head **12** is mounted on the tip of the support member **14***a*.
[0031] The SPM **13** is driven by a driving current (or driving voltage) supplied by the driver IC **16**.
[0032] The support member **14***a* supports the magnetic head **12** and is provided at the tip of the arm **15***a*. The support member **14***a* is, for example, a slider or a suspension.
[0033] The MA **14** is attached to the base or tip of the support member **14***a* and drives the support member **14***a*. The MA **14** is a

piezoelectric device such as a piezoelectric element. The MA **14** corresponds to an actuator that drives the tip side of a two-stage actuator. The MA **14** can be driven faster and more accurately than the VCM **15**. Therefore, the seek control can be sped up by operating the MA **14** prior to the VCM **15**.
[0034] The VCM **15** rotates the arm **15***a*. The VCM **15** corresponds to an actuator that moves the head relatively largely. In contrast, the MA **14** corresponds to an actuator that moves the support member **14***a* provided at the tip of the arm **15***a* slightly (in a relatively small manner).
[0035] The VCM **15** and the MA **14** are driven by an input control signal supplied by the driver IC **16**. This causes the magnetic head **12** to move toward the radial direction of the magnetic disk **11**. The input control signal is, for example, a driving current or a driving voltage. In other words, the magnetic head **12** is an example of a control target. The VCM **15** and the MA **14** are also examples of the control targets.
[0036] The driver IC **16** drives the SPM **13**, the MA **14**, and the VCM **15** according to control by a central processing unit (CPU) **186** to be described below in the system LSI **18**.
[0037] The head IC **17** amplifies a signal (read signal) read by the magnetic head **12**. The head IC **17** converts a write data transferred from a read/write (R/W) channel **181** to be described below in the system LSI **18** into a write current to be output to the magnetic head **12**.
[0038] The system LSI **18** is an LSI called a system on a chip (SoC) in which a plurality of elements are integrated on one chip. The system LSI **18** includes the R/W channel **181**, a hard disk controller (HDC) **182**, a buffer random access memory (RAM) **183**, a flash memory **184**, a program read only memory (ROM) **185**, the CPU **186**, and a RAM **187**.
[0039] The R/W channel **181** is a signal processing device that processes a group of signals related to reading and writing. The R/W channel **181** digitizes the read signal and decodes the read data from the digitized data. The R/W channel **181** also acquires, from the digitized data, a servo data necessary for positioning the magnetic head **12**. Furthermore, the R/W channel **181** encodes the write data.
[0040] The HDC **182** is connected to the host **20** via the host interface IF. The HDC **182** receives commands (write commands, read commands, etc.) transferred from the host **20**. The HDC **182** controls data transfer between the host **20** and the HDC **182**. The HDC **182** also controls data transfer between the magnetic disk **11** and the HDC **182**.
[0041] The buffer RAM **183** constitutes a buffer area for temporarily storing data to be written to and read from the magnetic disk **11** via the head IC **17** and the R/W channel **181**.
[0042] The flash memory **184** is a rewritable nonvolatile memory.
[0043] The program ROM **185** stores a control program (firmware). The control program may be stored in some areas of the flash memory **184**. The control program is a computer program used after shipment.
[0044] The CPU **186** functions as a main controller of the HDD **10**. The CPU **186** controls at least some elements in the HDD **10** according to the control program or an adjustment program stored in the program ROM **185**. The CPU **186** includes a seek control section **19** described below.
[0045] At least some areas of the RAM **187** is used as a working area for the CPU **186**.
[0046] The HDD **10** configured as described above includes a feedback control system that performs the seek control of the magnetic head **12**. The seek control refers to control the positioning of the magnetic head **12** at a target track. The feedback control system is executed, for example, by the CPU **186**, at regular time intervals, that is, each time servo data is acquired. Hereafter, this regular time interval is referred to as a sampling time or a sampling period. A step of sampling data at each sampling time is referred to as a sampling step.
[0047] FIG. **2** is a block diagram illustrating an example configuration of a seek control section **19**. The seek control section **19** executes seek control of the magnetic head **12**.
[0048] In the seek control section **19**, a feedback path F is used for each of the VCM **15** and the MA **14**. In the present embodiment, the learning control is assumed to act on the movement of the VCM **15**, and a configuration of the feedback path F with respect to the VCM **15** is described as an example.
[0049] The seek control section **19** is a digital control device that performs learning control of iteratively controlling a control target **34** and sequentially updating learning control inputs to improve control performance at each iteration.
[0050] The control target **34** has an actuator and is controlled by the seek control section **19**. The control target **34** is a target of state control by the seek control section **19**, and is the magnetic head **12** to be position-controlled by the VCM **15** and the MA **14** in the HDD **10**, semiconductor manufacturing equipment, a robot, and the like. The control target **34** may have the position of an end effector of an arm and the angle of each joint controlled.
[0051] A state of the control target **34** is, for example, a head position of the magnetic head **12** on the magnetic disk **11**, the position of the robot, and the like. The state of the control target **34** is not limited to position. For example, the state of the control target **34** may be a combination of a position and a velocity, a combination of the velocity and an acceleration, a combination of the position, the velocity, the acceleration, and external forces. The state of the control target **34** preferably includes at least one of the position and the velocity of the control target **34**. The state of the control target **34** may include the external forces applied to the control target **34**. The external forces applied to the control target **34** are, for example, bias forces.
[0052] In the present embodiment, an example in which the control target **34** is the magnetic heads **12** to be position-controlled by the VCM **15** and the state of the control target **34** is the head position of the magnetic head **12** will be described.
[0053] The seek control section **19** includes a learning control unit **40**, a feedback control unit **30**, a notch filter **31**, an adder **33**, the VCM **15**, an error calculation unit **35**, and an adder **36**.
[0054] The VCM **15** operates according to the input control signal sequentially received from the feedback control unit **30** via the notch filter **31** and the adder **33**, and sequentially outputs an operation result state representing a state of an operation result. The input control signal is, for example, the driving current or the driving voltage, as described above.
[0055] In the present embodiment, the magnetic head **12** sequentially outputs, as the operation result state, the position of the magnetic head **12** position-controlled by the VCM **15** on the magnetic disk **11**. The operation result state may be configured to be

detected by a detection device such as a known sensor or based on servo data acquired via the magnetic head **12**.
[0056] The error calculation unit **35** calculates a tracking error. The tracking error represents an error of the operation result state of the control target **34** relative to a target state. In other words, the tracking error represents an error of a current state of the control target **34** relative to the target state. In the present embodiment, the error calculation unit **35** calculates a position error between the head position of the magnetic head **12** and a target trajectory as the tracking error. The target trajectory is a target position of the magnetic head **12** and is an example of the target state.
[0057] The error calculation unit **35** outputs the calculated position error to the learning control unit **40** and the feedback control unit **30**. In other words, the error calculation unit **35** sequentially receives the head position which is the operation result state output at each sampling period, calculates a position error which is the tracking error relative to the target trajectory each time the head position is received, and outputs the position error to the learning control unit **40** and the feedback control unit **30**.
[0058] The adder **36** adds the position error received from the error calculation unit **35** to the learning control input received from the learning control unit **40**, and outputs the result to the feedback control unit **30**.
[0059] The learning control input is a learning value that is learned through iterative learning by the learning control unit **40**. The learning control input is used to correct a signal output to the control target **34** such as the VCM **15**. In other words, the learning control input represents a correction amount used during learning control.
[0060] The feedback control unit **30** outputs a feedback signal to cause the operation result state of the control target **34** to track the target state, based on an input signal according to the tracking error between the operation result state of the control target **34** and the target state. In the present embodiment, the feedback control unit **30** outputs the feedback signal to cause the head position of the magnetic head **12**, the head position being moved according to the input control signal, to track the target trajectory, based on the position error of the head position relative to the target trajectory on the magnetic disk **11**.
[0061] The notch filter **31** is a filter used to stabilize mechanical resonance of the VCM **15**. The notch filter **31** removes mechanical resonance frequency components of the VCM **15** from the feedback signal received from the feedback control unit **30** using the filter, and outputs a resultant signal to the adder **33**.
[0062] The adder **33** outputs the calculation result of adding the output signal of the notch filter **31** and the inputted feed-forward (FF) input **32** to the VCM **15** as the above input control signal. The FF input **32** represents an input for feed-forward control of the VCM **15**. The FF input **32** is expressed, for example, in the dimension of acceleration.
[0063] Thus, the seek control section **19** is provided with the feedback path F. The feedback path F is a communication path through which the input control signal, according to the feedback signal output by the feedback control unit **30**, is input to the VCM **15** (control target **34**) and also the tracking error (position error) based on the operation result state (head position) of the control target **34** (magnetic head **12**) according to the input control signal is input to the feedback control unit **30**. The feedback path F operates such that the position error between the head position of the magnetic head **12**, which is output by the VCM **15**, and the target trajectory of the seek control approaches zero asymptotically.
[0064] The learning control unit **40** outputs to the feedback path F, through which the input signal according to the tracking error is sent to be input to the feedback control unit **30**, the learning control input that has been updated according to the tracking error to cause the tracking error to approach zero asymptotically. Since the seek control section **19** includes the learning control unit **40**, seek settling can be improved.
[0065] In the present embodiment, the learning control unit **40** outputs to the feedback path F, through which an input signal according to the position error is sent to be input to the feedback control unit **30**, the learning control input updated according to the position error to cause the position error to approach zero asymptotically. Furthermore, in the present embodiment, an example in which the learning control unit **40** outputs the learning control input to an input end of the feedback control unit **30** in the feedback path F will be described. Therefore, in the present embodiment, the adder **36** outputs an addition result obtained by adding the position error to the learning control input to the feedback control unit **30** as an input signal.
[0066] The learning control unit **40** may output the learning control input to the input end of the VCM **15**, which is the control target **34** in the feedback path F. In this case, the adder **36** is arranged on the input end of the VCM, and an addition result obtained by adding the learning control signal to a calculation result obtained by adding the output signal of the notch filter **31** to the FF input **32** using the adder **33** may be output to the VCM **15** as the input control signal. As described above, the control target **34** of the seek control section **19** may be semiconductor manufacturing equipment, a robot, and the like. For example, if the control target **34** is a robot, the feedback path F may be applied to the position of an end effector of the robot or to the angle of each joint instead of the head position of the magnetic head **12**.
[0067] Hereinafter, a conventional learning control will be described.
[0068] FIG. **3** is an explanatory diagram of an example of conventional learning control. As illustrated in FIG. **3**, in the conventional learning control, a head positioning accuracy of the magnetic head **12** is improved in a section (period) where the learning control inputs generated by the learning control and the target trajectory are applied to the feedback path F. However, in the conventional learning control, a transient response may occur at the moment when the learning control is terminated, and head positioning accuracy may deteriorate. In other words, in the conventional technologies, the accuracy of the operation result state of the control target **34** relative to the target state may decrease.
[0069] First, the reason why the transient response occurs at the end of learning control will be described in detail.
[0070] FIG. **4** is a schematic diagram of an example of the seek control section **19**.
[0071] To analyze an operation of the learning control, it is assumed that each element of the seek control section **19** described using FIG. **2** is a linear system, and a time response of the position error during the seek control when the learning control is active will be considered. This consideration uses a sequence of impulse responses p.sub.1, p.sub.2, . . . , p.sub.n-1, p.sub.n of the transfer characteristics of a closed loop from e to u, which connects the learning control unit **40** in the feedback path F as illustrated in FIG. **4**. e represents the tracking error. In other words, in the present embodiment, e represents the position error. u represents the learning control input. The sequence of impulse responses is a group of position errors e at each sampling time output from the control target **34** over time when one learning control input u is output from the learning control unit **40** to the

feedback path F. n is an integer of 2 or more. Numerical subscripts of P represent the sampling times i=0, 1, 2, . . . , n.

[0072] Let e be a vector in which the above position errors are arranged at each sampling time, P be a matrix in which the above sequences of impulse responses p.sub.1, p.sub.2, . . . , p.sub.n-1, p.sub.n are arranged in lower triangle portion in expression (2) described below while being shifted by one sampling time, u be a vector in which each of the learning control inputs generated by the learning control using the learning control unit **40** is arranged at each corresponding sampling time, and d be a vector of position errors in the seek control when the learning control is disabled, then the following expression (1) is established.

[00001] $e = \mathrm{P}u + d$ expression(1)

[0073] In expression (1), e represents the vector of position errors, P represents the matrix in which sequences of impulse responses p.sub.1, p.sub.2, . . . , p.sub.n-1, p.sub.n are arranged while being shifted by one sampling time, u represents the vector of learning control inputs, and d represents the vector of position errors when the learning control is disabled. In the following description, the vector of position errors may be referred to as a position error vector and the vector of learning control inputs as a learning control input vector.

[0074] For example, when an output section length of an output section of the learning control input output from the learning control unit **40** to the feedback path F is “7” and components of the above expression (1) are expressed at each sampling time i=0, 1, 2, . . . , n, expression (2) is obtained.

[00002]

$$
\begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \\ e_5 \\ e_6 \\ e_7 \\ e_8 \\ e_9 \\ e_{10} \\ \vdots \\ e_n \end{bmatrix}
=
\begin{bmatrix}
p_1 & & & & & & & & & & & \\
p_2 & p_1 & & & & & & & & & & \\
p_3 & p_2 & p_1 & & & & & & & & & \\
p_4 & p_3 & p_2 & p_1 & & & & & & & & \\
p_5 & p_4 & p_3 & p_2 & p_1 & & & & & & & \\
p_6 & p_5 & p_4 & p_3 & p_2 & p_1 & & & & & & \\
p_7 & p_6 & p_5 & p_4 & p_3 & p_2 & p_1 & & & & & \\
p_8 & p_7 & p_6 & p_5 & p_4 & p_3 & p_2 & p_1 & & & & \\
p_9 & p_8 & p_7 & p_6 & p_5 & p_4 & p_3 & p_2 & p_1 & & & \\
p_{10} & p_9 & p_8 & p_7 & p_6 & p_5 & p_4 & p_3 & p_2 & p_1 & & \\
\vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \\
p_n & p_{n-1} & p_{n-2} & p_{n-3} & p_{n-4} & p_{n-5} & \vdots & p_5 & p_4 & p_3 & p_2 & p_1
\end{bmatrix}
\begin{bmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}
+
\begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ d_6 \\ d_7 \\ d_8 \\ d_9 \\ d_{10} \\ \vdots \\ d_n \end{bmatrix}
\quad \text{expression(2)}
$$

[0075] In general learning control, the learning control unit **40** stores the learning control inputs in a memory provided in the learning control unit **40** at each seek control and iteratively updates the learning control inputs based on the position error in the next seek control.

[0076] FIGS. **5** and **6** are explanatory diagrams of examples of the conventional learning control.

[0077] In the conventional technologies, when the learning control inputs stored in the memory are updated, position error signals e.sub.1 to e.sub.7 in an evaluation section having the same length as the length of the output section of the learning control inputs output from the learning control unit to the feedback path F are used. Therefore, when the learning converges sufficiently, learning control inputs converge to expression (3), as illustrated in FIG. **5**, where P.sub.sq and d.sub.sq are portions at which the learning control inputs u.sub.0 to u.sub.6 act on the matrix P and the vector d.

[00003] $u_{\mathrm{sq}} = -P_{\mathrm{sa}}^{-1} d_{\mathrm{sq}}$ expression(3)

[0078] As illustrated in FIG. **6**, the learning control inputs u.sub.sq after convergence in the conventional technology, which are determined by expression (3), are zero in the range of the position errors e.sub.0 to e.sub.7, and therefore, the position errors are theoretically completely suppressed. However, in practice, the position errors e.sub.8, . . . after the position error e.sub.7 do not necessarily become zero, and this influence appears as a transient response at the end of the learning control between the position error e.sub.7 and the position error e.sub.8.

[0079] Thus, in the conventional technology, a transient response may occur at the moment when the learning control is terminated, and head positioning accuracy may deteriorate. In other words, in the conventional technology, the accuracy of the operation result state of the control target **34** relative to the target state may decrease.

[0080] Therefore, the learning control unit **40** according to the present embodiment controls an evaluation section length N of the evaluation section in which the position error is evaluated, which is the tracking error, to be longer than an output section length M of the output section in which the learning control unit **40** outputs the learning control inputs to the feedback path F.

[0081] FIG. **7** is a schematic diagram illustrating an example of the relationship between the evaluation section length N of the evaluation section S**1** and the output section length M of the output section S**2** in the conventional technology. In FIG. **7**, the black dots indicate the sampling points for each of the position errors and the learning control inputs. As illustrated in FIG. **7**, in the conventional technology, the evaluation section length N matches the output section length M of the output section S**2**. Furthermore, in the conventional technology, the number of samplings in the evaluation section S**1** matches the number of outputs in the output section S**2**.

[0082] FIG. **8** is a schematic diagram illustrating an example of the relationship between the evaluation section length N of the evaluation section S**1** and the output section length M of the output section S**2** according to the present embodiment. In FIG. **8**, the black dots indicate the sampling points for each of the position errors and the learning control inputs. As illustrated in FIG. **8**, in the present embodiment, the evaluation section length N of the position errors is longer than the output section length M of the learning control inputs. In the present embodiment, the number of outputs in the output section S**2** is less than the number of samplings in the evaluation section S**1**. In the present embodiment, the timing of the last sampling e.sub.10 in the evaluation

section S**1** is located later than the timing of the last output us in the output section S**2**. In the present embodiment, for example, the time interval of sampling in the evaluation section S**1** matches the time interval of output in the output section S**2**.
[0083] As illustrated in FIG. **8**, the evaluation section S**1** and the output section S**2** partially overlap each other. The evaluation section S**1** includes a timing period later (later in time) than the output section S**2**. In detail, as illustrated in FIG. **8**, the start timing of evaluation section S**1** matches the start timing of output section S**2**, and the end timing of evaluation section S**1** is later than the end timing of output section S**2**.
[0084] In other words, in the present embodiment, in order to mitigate the transient response, the learning control unit **40** performs iterative learning including the position errors e.sub.8 to e.sub.n after the learning control inputs u.sub.0 to u.sub.6 have been applied to the feedback path F so as to satisfy the relationship of the evaluation section length N>the output section length M.
[0085] Therefore, as illustrated in FIG. **9**, the learning control unit **40** extends portions at which the learning control inputs u.sub.0 to u.sub.6 act on the matrix P and the vector d as compared with the conventional technologies (see FIG. **5**). FIG. **9** is an explanatory diagram of an example of learning control according to the present application. The learning control unit **40** performs control such that the evaluation section length N of the position errors is longer than the output section length M of the learning control inputs by performing iterative learning for obtaining the least-squares solution of the following expression (4) with the matrix P after extension and the vector d after extension defined as P.sub.ls and d.sub.ls, respectively.
[00004] $u_{ls} = -P_{ls}^{\text{Math.}} d_{ls}$ expression(4) $\dagger$ expression(4A)
[0086] In expression (4), u.sub.ls represents the vector of learning control inputs, p.sub.ls represents the matrix in which the sequences of impulse responses are arranged while being shifted by one sampling time, dis represents the vector of tracking errors when the learning control is disabled, and expression (4A) in the expression (4) represents a pseudo-inverse matrix operation.
[0087] According to the learning control inputs u.sub.ls in expression (4), the position error signals e do not converge to zero completely in principle. However, unlike the conventional learning control, expression (4) also takes into account the position errors after the learning control inputs have been applied to the feedback path F. Therefore, the learning control unit **40** can perform control such that the evaluation section length N of the position errors is longer than the output section length M of the learning control inputs by performing iterative learning for obtaining the least-squares solution of expression (4), and can thereby mitigate the above transient response.
[0088] Furthermore, it is considered that the learning control unit **40** iteratively updates the learning control inputs at each seek control by performing the iterative learning according to expression (4). In this case, since expression (4) is the least-squares solution, in iterative updating the learning control inputs, u.sub.ls in expression (4) is approximately obtained by performing iterative calculation using a gradient method with an objective function that minimizes a squared norm of the position error vector e.sub.ls. The update expression for this iterative calculation is expressed by expression (5) below.
[00005] $u_{ls}^{(j+1)} = u_{ls}^{(j)} + \quad P_{is}^{T} e_{ls}^{(j)}$ expression(5)
[0089] In expression (5), u.sub.ls represents the vector of learning control inputs, pis represents the matrix in which the sequences of impulse responses are arranged while being shifted by one sampling time, and e.sub.ls represents the vector of the position errors. In expression (5), j represents the number of learning iterations, and β represents the learning gain.
[0090] In other words, the learning control unit **40** can perform control such that the evaluation section length N of the position errors is longer than the output section length M of the learning control inputs by performing the iterative learning according to expression (5).
[0091] Here, the matrix P.sub.ls including the sequences of impulse responses of the closed-loop transfer characteristics in expression (5) may be difficult to acquire accurately in practice. Therefore, some approximation is needed. Consequently, the learning control unit **40** calculates the closed-loop transfer characteristic model based on a model of the VCM **15** to acquire its impulse response.
[0092] FIG. **10** illustrates an example of the impulse responses when the closed-loop transfer characteristic model is calculated based on the model of the VCM **15**.
[0093] The learning control unit **40** truncates the acquired impulse responses at a length m that can roughly approximate a waveform of the impulse response, and extracts the truncated impulse responses (see FIG. **10**). Then, in the learning control unit **40**, an approximate matrix of P.sub.ls is configured as in the following expression (6) using the extracted sequence of impulse responses p.sub.1, p.sub.2, . . . , p.sub.m-1, p.sub.m. Consequently, the learning control unit **40** can perform control such that the evaluation section length N of the position errors is longer than the output section length M of the learning control inputs by performing the iterative learning according to expression (7).

[00006] $$\tilde{P}_{ls} = \left[\begin{array}{cccccc} p_1 & & & & & \\ p_2 & p_1 & & 0 & & \\ \text{.Math.} & p_2 & \ddots & & & \\ p_{m-1} & \text{.Math.} & \ddots & p_1 & & \\ p_m & p_{m-1} & \ddots & p_2 & p_1 & \\ & p_m & \ddots & \text{.Math.} & p_2 & \\ & & \ddots & p_{m-1} & \text{.Math.} & \\ & 0 & & p_m & p_{m-1} & \\ & & & & p_m & \end{array} \right] \text{ expression(6)} \quad \tilde{P}_{ls} \text{ expression(6}A\text{)}$$

[0094] Expression (6A) as the left-hand side in the expression (6) represents an approximation matrix of P.sub.ls in expression

(5). In expression (6), p.sub.1 to p.sub.m represent a sequence of m impulse responses.

[0095] The update expression for learning using the approximation matrix of P.sub.ls is expressed by expression (7) below.

[00007] $u_{ls}^{(j+1)} = u_{ls}^{(j)} + \tilde{P}_{ls}^{T} e_{ls}^{(j)}$ expression(7)

[0096] In expression (7), u.sub.ls represents the vector of learning control inputs and e.sub.ls represents the vector of position errors. In expression (7), j represents the number of learning iterations, and β represents the learning gain. Expression (6A) in the expression (7) represents expression (6).

[00008] $\tilde{P}_{ls}^{T} e_{ls}^{(j)}$ expression(7A)

[0097] Here, expression (7A) in the expression (7) is an operation of a matrix×a vector. This operation may be difficult to be performed with the computing capability of the CPU **186** (see FIG. **1**) of the actual HDD **10**.

[0098] Therefore, in the learning control unit **40**, it is preferable that a calculation of expression (7A) is replaced with a Finite Impulse Response (FIR) filter.

[0099] For this FIR filter, a filter expressed by the following expression (8) with the coefficients of the sequence of impulse responses p.sub.1, p.sub.2, . . . , p.sub.m-1, p.sub.m extracted in FIG. **10** being arranged in reverse time order may be used.

[00009] $F(z) = p_{m} + p_{m-1} z^{-1} + .Math. + p_{2} z^{m-2} + p_{1} z^{m-1}$ expression(8)

[0100] In expression (8), F (z) represents the filter and z represents a delay operator. In expression (8), p.sub.1 to p.sub.m represent a sequence of m impulse responses.

[0101] Let y be a vector output when the position error vector e.sub.ls.sup.(j) is input to the FIR filter expressed by expression (8), as illustrated in FIG. **11**, the remainder obtained by removing m−1 delayed components of the FIR filter from y is equal to the vector obtained by the calculation of the above expression (7A) in the above expression (7). FIG. **11** is an explanatory diagram of the FIR filter.

[0102] FIG. **12** is a schematic diagram of an example of a configuration of the learning control unit **40** according to the present embodiment, including the FIR filter.

[0103] The learning control unit **40** includes an FIR filter **41**, a gain multiplier **42**, an adder **43**, and a memory **44**.

[0104] A memory **44** is used to store the learning control input at each sampling step i. A memory length of the memory **44** may be the same length as the output section length M.

[0105] The FIR filter **41** outputs, to the gain multiplier **42**, the position error signal after filtering the position error e.sub.ls.sup.(j)[i] input by the feedback path F at each sampling step i. The gain multiplier **42** multiplies the position error signal received from the FIR filter **41** by a gain β. The adder **43** updates the learning control input stored in a memory at a sampling step i−m−d+1 among memories included in the memory **44** to an addition result obtained by adding a multiplication result obtained by multiplying the position error signal by the gain β to the learning control input read from the memory **44**. m represents a FIR filter length. m−d+1 corresponds to the phase delay amounts of both a phase delay of the control target **34**, which exhibits the closed-loop transfer characteristic illustrated in FIG. **4**, and a phase delay of the FIR filter **41**.

[0106] Therefore, the learning control input stored in the memory **44** at the sampling step i−m−d+1 going back in time from the current time i is updated according to a newly observed position error.

[0107] Thus, in the present embodiment, in order to prevent the learning from becoming unstable due to both the phase delay of the control target **34**, which exhibits the closed-loop transfer characteristic illustrated in FIG. **4**, and the phase delay of the FIR filter **41**, the learning control unit **40** updates the memory backward from the current time i by both the phase delay amounts. In other words, the time at which the learning control unit **40** reads a value stored in the memory **44** from the memory **44** as the learning control input to be output to the feedback path F is a time in the future relative to the time when the memory **44** is updated. Therefore, the learning control unit **40** can advance the phase by updating the memory backward from the current time i by both the above phase delay amounts, and can thereby correct the above phase delays.

[0108] As described above, the HDD **10** (learning control device, magnetic disk device) according to the present embodiment includes the feedback control unit **30** and the learning control unit **40**.

[0109] The feedback control unit **30** outputs, based on an input signal according to a tracking error (position error) between an operation result state of the control target **34** operating according to an input control signal based on a feedback signal and a target state, the feedback signal to cause the operation result state (head position) of the control target **34** (magnetic head **12**) to track the target state (target trajectory). The learning control unit **40** outputs to the feedback path F, through which the input signal according to the tracking error is sent to be input to the feedback control unit **30**, the learning control input that has been updated according to the tracking error to cause the tracking error to approach zero asymptotically. The evaluation section length N of the evaluation section S**1** of the tracking error by the learning control unit **40** is longer than the output section length M of the output section S**2** in which the learning control unit **40** outputs the learning control inputs to the feedback path F (N>M).

[0110] Therefore, the learning control unit **40** according to the present embodiment can suppress the transient response occurring at the moment when the learning control is terminated, which occurs in the conventional technology, and can suppress the accuracy deterioration of the operation result state of the control target relative to the target state.

[0111] Consequently, the HDD **10** (learning control device, magnetic disk device) according to the present embodiment can improve the accuracy of the operation result state of the control target **34** relative to the target state.

[0112] When the HDD **10** according to the present embodiment is a magnetic disk device, the positioning accuracy of the head position of the magnetic head **12** can be improved.

Effects

[0113] FIGS. **13** to **19** are explanatory diagrams of results of numerical simulations of the HDD **10** according to the present embodiment and the conventional technologies.

[0114] FIGS. **13**, **14**, and **15** illustrate Bode plots of a model of the VCM **15**, the notch filter **31**, and the feedback control unit **30** illustrated in FIG. **2**, respectively. FIGS. **16** and **17** illustrate the time histories of the target trajectory and the FF input **32** in FIG.

2, respectively. The model of the VCM **15**, the notch filter **31**, and the feedback control unit **30** illustrated in the reference "http://www2.iee.or.jp/~dmec/committee/DMEC1005/dsa_HDD_ben ch_e.html" (HDD benchmark problem) were used.
[0115] FIG. **18** is a diagram comparing the numerical simulation results of operations of the conventional learning control using the above components, and FIG. **19** is a diagram comparing the numerical simulation results of operations of the learning control according to the present embodiment using the above components. FIG. **18** illustrates the numerical simulation results of operations of the conventional learning control. FIG. **19** illustrates the numerical simulation results of operations of the learning control according to the present embodiment.
[0116] In both FIG. **18** and FIG. **19**, the output section length M of the output section S**2** of the learning control inputs is a length of 48 samples. The evaluation section length N of the evaluation section S**1** of the position errors in the conventional learning control illustrated in FIG. **18** is a length of 48 samples. The evaluation section length N of the evaluation section S**1** of the position errors in the learning control according to the present embodiment illustrated in FIG. **19** is a length of 78 samples. In other words, in the learning control according to the present embodiment illustrated in FIG. **19**, in order to mitigate the transient response, the length of the position error signals included in the learning after the end of the learning control was set to 30 samples.
[0117] FIGS. **18** and **19** each illustrate curve diagrams representing both the head position trajectory in the seek control without the learning control and the head position trajectory after the learning has converged sufficiently. Due to the effect of the learning control, the head position tracks the target trajectory indicated by the dashed line in the figures well in the section where the learning control inputs are applied to the feedback path F in both learning. However, in the conventional learning control illustrated in FIG. **18**, a transient response occurs at the moment when the application of the learning control input is terminated, and the head position is offset. On the other hand, it is found that this transient response is mitigated in the learning control according to the present embodiment illustrated in FIG. **19**.
[0118] As described above, the present embodiment can suppress the transient response occurring at the moment when the learning control is terminated, which occurs in the conventional technology, and can suppress the accuracy deterioration of the operation result state of the control target relative to the target state. Consequently, the HDD **10** (learning control device, magnetic disk device) according to the present embodiment can improve the accuracy of the operation result state of the control target **34** relative to the target state. When the HDD **10** according to the present embodiment is a magnetic disk device, the positioning accuracy of the head position of the magnetic head **12** can be improved.
[0119] Each unit of the seek control section **19** illustrated in FIG. **2** is implemented by, for example, one or more processors. For example, each of the above units may be implemented by causing a processor such as the CPU **186** to execute a computer program, that is, by software. Each of the above units may be implemented by a processor such as a dedicated integrated circuit (IC), that is, by hardware. Each of the above units may be implemented using a combination of software and hardware. When a plurality of processors are used, each processor may implement one of the units or two or more of the units.
[0120] A computer program to be executed by the magnetic disk drive and the learning control device according to the embodiment is installed in a program ROM or the like in advance and provided.
[0121] The computer program to be executed by the magnetic disk device and the learning control device according to the embodiment may be an installable or executable format file and stored in a computer readable storage medium such as a compact disc read only memory (CD-ROM), a flexible disk (FD), a compact disc-recordable (CD-R), and a digital versatile disc (DVD), and may be provided as a computer program product.
[0122] Furthermore, the computer program to be executed by the magnetic disk device and the learning control device according the embodiment may be stored in a computer connected to a network such as the Internet, and may be provided by having the computer program downloaded via the network. The computer program to be executed by the magnetic disk drive and the learning control device according to the embodiment may be configured to be provided or distributed via a network such as the Internet.
[0123] The computer program to be executed by the magnetic disk device and the learning control device according to the embodiment can cause the computer to function as each unit of the magnetic disk device described above. In this computer, the CPU **186** can read a computer program from a computer readable storage medium onto a main memory to execute the computer program.
[0124] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the inventions. Indeed, the novel embodiments described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the embodiments described herein may be made without departing from the spirit of the inventions. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the inventions.

Claims

1. A learning control device comprising: one or more hardware processors configured to function as: a feedback control unit that outputs, based on an input signal according to a tracking error between an operation result state of a control target operating according to an input control signal based on a feedback signal and a target state, the feedback signal to cause the operation result state of the control target to track the target state; and a learning control unit that outputs to a feedback path, through which the input signal according to the tracking error is sent to be input to the feedback control unit, a learning control input updated according to the tracking error to cause the tracking error to approach zero asymptotically, wherein an evaluation section length of an evaluation section by the learning control unit for the tracking error is longer than an output section length of an output section in which the learning control unit outputs the learning control input to the feedback path.
2. The learning control device according to claim 1, wherein the learning control unit performs iterative learning for obtaining a

least-squares solution of expression (4), $u_{\mathrm{ls}} = -P_{\mathrm{ls}}^{\dagger} d_{\mathrm{ls}}$ expression(4) † expression(4A) where, in expression (4), u.sub.ls represents a vector of learning control inputs, P.sub.ls represents a matrix in which sequences of impulse responses are arranged while being shifted by one sampling time, d.sub.ls represents a vector of tracking errors when learning control is disabled, and expression (4A) in expression (4) represents a pseudo-inverse matrix operation.

3. The learning control device according to claim 2, wherein the learning control unit performs iterative learning according to expression (5), $u_{\mathrm{ls}}^{(j+1)} = u_{\mathrm{ls}}^{(j)} + P_{\mathrm{is}}^{T} e_{\mathrm{ls}}^{(j)}$ expression(5) where, in expression (5), u.sub.ls represents a vector of learning control inputs, P.sub.ls represents a matrix in which sequences of impulse responses are arranged while being shifted by one sampling time, e.sub.ls represents a vector of position errors, j represents a number of learning iterations, and β represents a learning gain.

4. The learning control device according to claim 3, wherein the learning control unit performs iterative learning according to expression (7),

$$\tilde{P}_{\mathrm{ls}} = \begin{bmatrix} p_1 & & & 0 & \\ p_2 & p_1 & & & \\ \text{.Math.} & p_2 & \ddots & & \\ p_{m-1} & \text{.Math.} & \ddots & p_1 & \\ p_m & p_{m-1} & \ddots & p_2 & p_1 \\ & p_m & \ddots & \text{.Math.} & p_2 \\ & & \ddots & p_{m-1} & \text{.Math.} \\ & 0 & & p_m & p_{m-1} \\ & & & & p_m \end{bmatrix} \quad \text{expression(6)} \quad \tilde{P}_{\mathrm{ls}} \quad \text{expression(6}A\text{)}$$

$u_{\mathrm{ls}}^{(j+1)} = u_{\mathrm{ls}}^{(j)} + \tilde{P}_{\mathrm{ls}}^{T} e_{\mathrm{ls}}^{(j)}$ expression(7) where, expression (6A) as a left-hand side in expression (6) represents an approximation matrix of P.sub.ls in expression (5); p.sub.1 to p.sub.m in expression (6) represent a sequence of m impulse responses; in expression (7), u.sub.ls represents a vector of learning control inputs, e.sub.ls represents a vector of position errors, j represents a number of learning iterations, and β represents a learning gain; and expression (6A) in the expression (7) represents expression (6).

5. The learning control device according to claim 4, wherein the learning control unit inputs a plurality of the tracking errors sequentially sampled from the feedback path to a filter expressed by expression (8), and updates the learning control input using the tracking errors output from the filter, $F(z) = p_m + p_{m-1} z^{-1} + \text{.Math.} + p_2 z^{m-2} + p_1 z^{m-1}$ expression(8) where, in expression (8), F (z) represents the filter, z represents a delay operator, and p.sub.1 to p.sub.m represent a sequence of m impulse responses.

6. The learning control device according to claim 1, wherein the evaluation section and the output section partially overlap each other.

7. The learning control device according to claim 1, wherein a start timing of the evaluation section is same as a start timing of the output section, and an end timing of the evaluation section is later than an end timing of the output section.

8. The learning control device according to claim 1, wherein the control target is a magnetic head, the operation result state is for a head position of the magnetic head on the magnetic disk, the target state is a target trajectory of the magnetic head, and the tracking error is a position error.

9. A learning control method performed by a computer of a learning control device, the method comprising: outputting, based on an input signal according to a tracking error between an operation result state of a control target operating according to an input control signal based on a feedback signal and a target state, the feedback signal to cause the operation result state of the control target to track the target state; and performing a learning control by outputting, to a feedback path, through which the input signal according to the tracking error is sent to be input to an outputting operation, a learning control input updated according to the tracking error to cause the tracking error to approach zero asymptotically, wherein an evaluation section length of an evaluation section by the learning control for the tracking error is longer than an output section length of an output section in which the learning control outputs the learning control input to the feedback path.

10. A magnetic disk device comprising: a feedback control unit that outputs, based on an input signal according to a position error of a head position of a magnetic head configured to move the head position according to an input control signal based on a feedback signal, relative to a target trajectory on a magnetic disk, the feedback signal to cause the head position of the magnetic head to track the target trajectory; and a learning control unit that outputs to a feedback path, through which the input signal according to the position error is sent to be input to the feedback control unit, a learning control input updated according to the position error to cause the position error to approach zero asymptotically, wherein an evaluation section length of an evaluation section by the learning control unit for the position error is longer than an output section length of an output section in which the learning control unit outputs the learning control input to the feedback path.